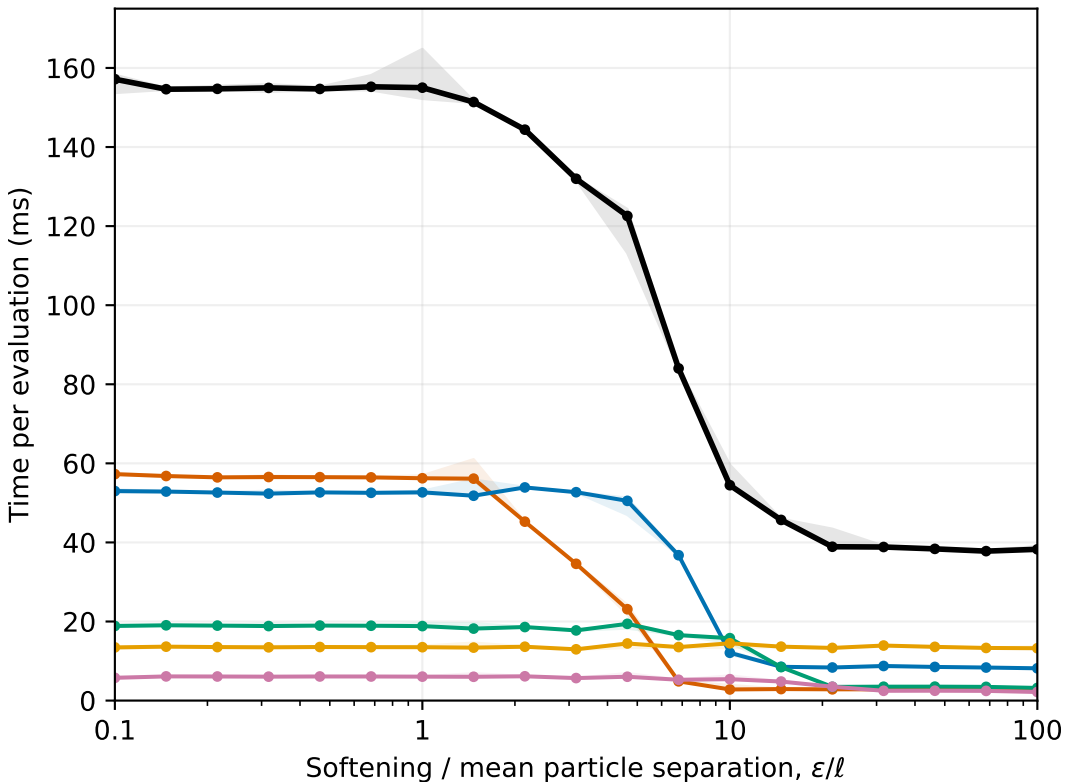
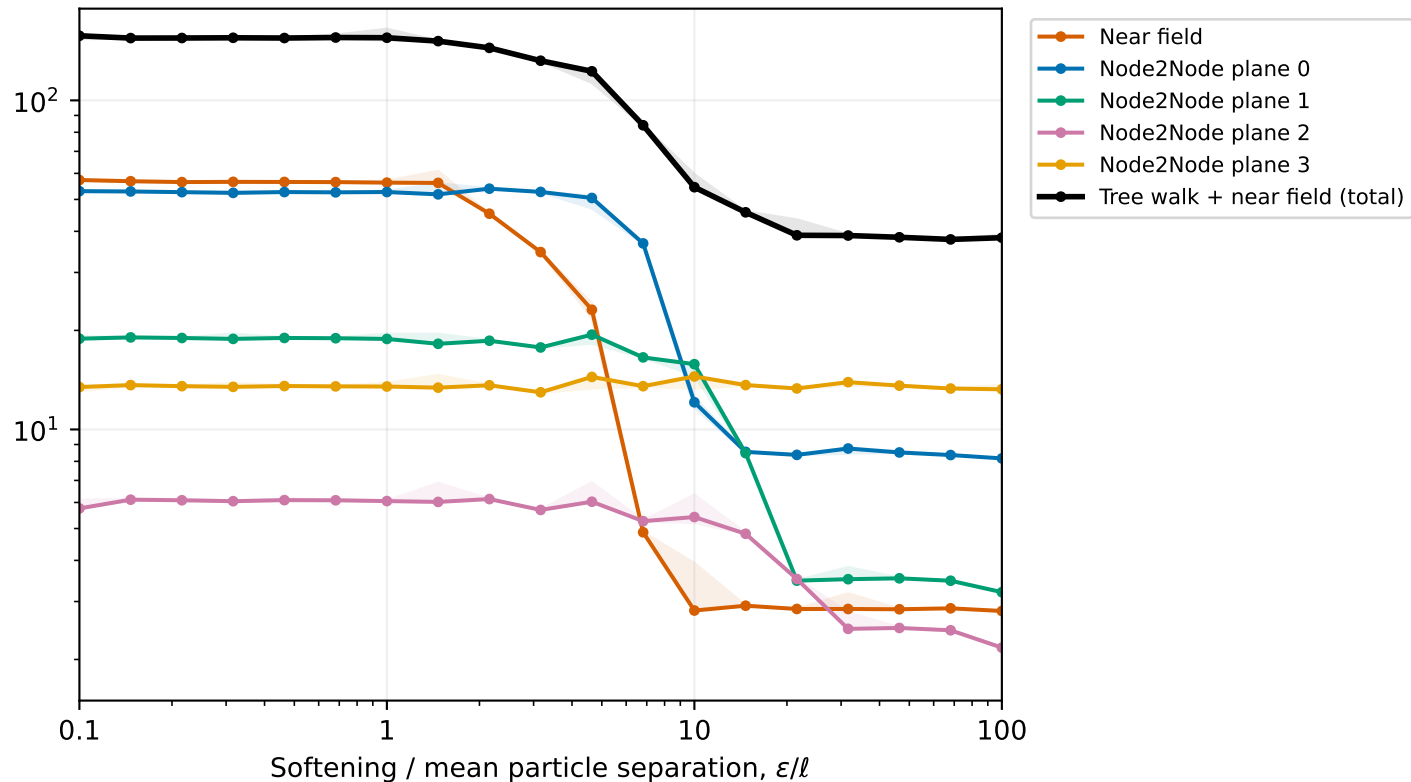


4 million uniform particles · GTX 1070 · softened opening ($\theta = 0.8$, $p = 5$)

Absolute cost



Logarithmic time scale



Total starts from a ready tree and multipoles; includes downward translations. Shading: observed min-max.